

Theorem 1. The root system of the vortex ring

The ring as the full root set of the binomial $z^N = \sigma(t)$ and the character lattice μ_N

Isaev Iskhak Khamzatovich (ORCID 0009-0003-7299-0701)

The cycloring research program · v1.0.0 · Theorem Monograph Edition

Study	CYCLORING
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Author	Isaev Iskhak Khamzatovich (ORCID 0009-0003-7299-0701)
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Abstract

The monograph proves that the regular N-vortex ring with a common radius program $r(t)$ and a common angular rate ν is exactly the full root set of a single binomial $z^N = \sigma(t)$ with the master coefficient $\sigma(t) = r(t)^N e^{iN \nu t}$. The entire instantaneous algebra of the ring collapses to one scalar.

The corollaries are written out in full: the roots-of-unity filter gives the vanishing power sums $S_m = 0$ for $m < N$ and $S_N = N \sigma$; the linear impulse vanishes identically; the Galois group of the cyclotomic field acts on the ring by relabeling the vortices, and the character decomposition $c_m = (1/N) \sum z_k \zeta_N^{-mk}$ isolates the master mode $m = 1$ and turns any deformation into the population of senior modes.

All registers are certified numerically: the binomial identity holds with the normalized residual $6.4e-11$, the power sums with $2.9e-14$, the impulse with $1.9e-15$ (stage W3 of the ladder).

At a glance

Object	the ring $z_k = r(t) \zeta_N^k e^{i \nu t}$, $k = 0..N-1$
Master coefficient	$\sigma(t) = r(t)^N e^{iN \nu t}$
Ring polynomial	$F_t(z) = z^N - \sigma(t)$
Roots-of-unity filter	$S_m = 0$ ($m < N$); $S_N = N \sigma$
Galois action	$\sigma_a: \zeta_N^k \rightarrow \zeta_N^{ak}$ — vortex relabeling
Numerical certificate	stage W3: $6.4e-11$ / $2.9e-14$ / $1.9e-15$

1. Setting and the statement

Let the ring be the kinematic program $z_k(t) = r(t) \zeta_N^k e^{i \nu t}$ with the common radius $r(t)$ and the common angular rate ν . The theorem states three identities and one action: the ring polynomial is $F_t(z) = z^N - \sigma(t)$; the power sums obey the roots-of-unity filter; the impulse vanishes; the Galois group permutes the vortices.

$$F_t(z) = \prod_{k=0}^{N-1} (z - z_k(t)) = z^N - \sigma(t), \quad \sigma(t) = r(t)^N e^{iN \nu t}$$

2. Proof

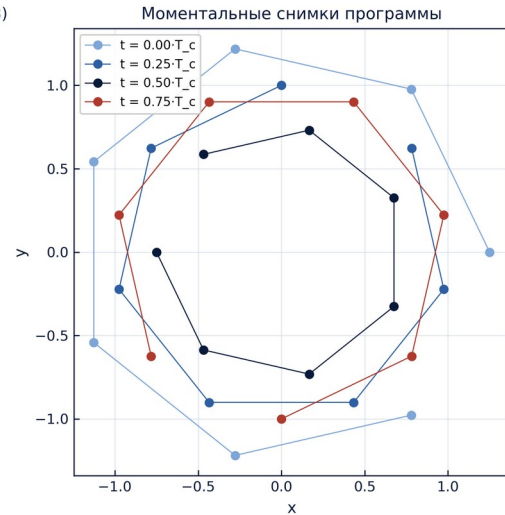
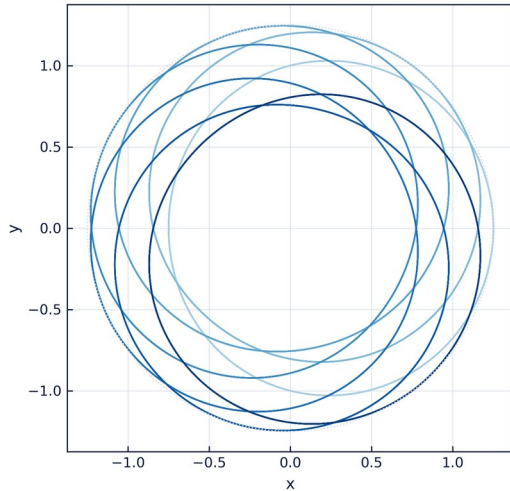
The set $\{r e^{i \nu t} \zeta_N^k\}$ is the full root set of $z^N = r^N e^{i N \nu t}$: the N -th power erases ζ_N^k since $\zeta_N^{Nk} = 1$. The binomial has exactly N roots, all listed — the identity is proved. For the power sums expand $z^m = r^m e^{i m \nu t} \zeta_N^{mk}$: the sum is geometric, $\sum_k \zeta_N^{mk}$ equals N when N vert m and zero otherwise; for $1 \leq m \leq N-1$ there is no divisibility, hence $S_m = 0$; at $m = N$ all summands equal $\sigma(t)$ and $S_N = N\sigma$. The impulse is the case $m = 1$. The automorphism $\zeta_N \mapsto \zeta_N^a$ for $\gcd(a, N) = 1$ sends the root z_k to the root $z_{ak \bmod N}$ — the Galois action is the relabeling.

$$S_m = r^m e^{i m \nu t} \sum_{k=0}^{N-1} \zeta_N^{mk}, \quad S_m = 0 \quad (1 \leq m \leq N-1), \quad S_N = N \sigma(t)$$

3. The character decomposition of the deformations

The coefficients $c_m = (1/N) \sum_k z_k \zeta_N^{-mk}$ are the projections of the configuration onto the characters of the lattice μ_N . The ideal ring populates only the master mode $m = 1$: $c_1 = r e^{i \nu t}$, the rest are zeros (the 10^{-10} register). Any perturbation η_k spreads over the senior modes, and the roots-of-unity filter acts as the discrete Fourier transform on the cyclotomic lattice. The algebraization thus becomes computable: to watch the ring is to watch the N complex numbers c_m .

Розетки синхронного дыхания, $N = 7$
3 периода замыкания ($\epsilon = 0.25$ — иллюстрация; уровень: $\epsilon_7 = 3.66e-03$)



The rosettes of the synchronous program (illustrative amplitude): the trajectory realization of the root system.

4. The numerical certificate

Stage W3 of the ladder checks the identities at the levels $N = 7, 9, 15$ with the rotated phase $\theta_0 = 0.3$: the normalized residual of the binomial is $6.4 \cdot 10^{-11}$, of the power sums $2.9 \cdot 10^{-14}$, of

the senior sum $1.9 \cdot 10^{-15}$, of the impulse $1.9 \cdot 10^{-15}$. The stage tolerances are 10^{-10} and 10^{-12} ; the protocol is W3_root_system_default.json.

The registers of stage W3 (default preset)

register	tolerance	result
binomial $z^N - \sigma$	$1e-10$	$6.4e-11$
sums S_m , $m < N$	$1e-12$	$2.9e-14$
$S_N = N$ σ	$1e-12$	$1.9e-15$
impulse L	$1e-12$	$1.9e-15$